Measuring each pass's blur and noise improves repeat-pass super-resolution of HiRISE: validation against rover and helicopter imagery at Jezero crater

Draft generated by `scripts/make_paper.py` from the result files in this repository (2026-11-04). Every number below is filled in from a results file; ***[pending]*** *marks a run that has not finished. Methods detail:* `docs/METHODS.md`*.*

Abstract

Repeat-pass super-resolution can sharpen orbital images of Mars. But such methods have been judged mostly by visual comparison and resolution claims, rarely against independent ground truth or with honest uncertainty.

We build on the Gotcha-PDE-TV (GPT) pipeline of Tao & Muller (2016). Their code is not public, so we re-implement a simplified version from the paper and add three measured ingredients:

- **Per-pass blur measured on a static target.** A compact, bright boulder at the landing area, visible in every pass, gives each pass's point-spread function. Blur varies 2.4× between the passes.
- **Noise-weighted fusion with robust pixel masks.** Each pass is weighted by its own white-noise level. The pixel masks reduce artefacts at a planted change in simulation; a real changed-ground test remains to be done.
- **A calibrated per-pixel uncertainty** from a pass-subset bootstrap.

On top of the measured restoration we add a confidence-scored detail layer. Rover-photo texture and rock exemplars are shown only above a threshold the user chooses, and are locked so that the result, seen through HiRISE's blur, is unchanged (lock residual ≤ 0.25%).

In simulation with exact truth (5 scenes), the extended pipeline lowers the 25 cm error from 3.55% to 1.43% of scene contrast against our GPT re-implementation (-60% relative). In simulation it also reduces the artefact at a planted change from 1.43 to 1.01. On real HiRISE the picture is mixed: GPT's independent pass halves agree slightly more than ours (mean coherence 0.263 vs 0.230, 25–100 cm), while our restoration of the boulder edge rings less than GPT's (11.7% vs 16.2% RMS; undershoot 26% vs 42%), at the cost of a wider edge (16.7 vs 14.1 cm). A centimetre-scale physical truth (Ingenuity's view of the parachute and backshell, placed in HiRISE by two independent methods) shows that, at any fixed regularisation, our restoration is much closer to the truth than the GPT re-implementation (50-25 cm error 0.206 against 0.333 at lambda = 0.04), and that with each method tuned to its own minimum the advantage is small (+0.023 in correlation in-sample, +0.009 out of sample), on one object.

The bootstrap uncertainty is conservative at the native 25 cm scale: held-out ±1σ coverage is 86%, against a nominal 68%, so it over-covers rather than under-covers. It is not reliable at the 6.25 cm output pixel.

Limits we measured:

- Detail that repeats between independent pass subsets reaches about 67 cm on real HiRISE and about 33 cm in simulation; the 6.3 cm output grid is a sampling choice, not a resolution

claim.
- Confidences are calibrated in simulation tuned to real pass-to-pass disagreement (expected calibration error 0.4%). The rover traverse truth is too poorly registered to check them below about 1 m (AUC 0.43–0.46).

Everything runs on a 4-core CPU.

1. Introduction

HiRISE images Mars at 25 cm per pixel (McEwen et al., 2007). Many sites, among them the Jezero crater landing site of Perseverance, have been imaged repeatedly. Multi-image super-resolution restoration (SRR) combines such passes into a finer image. Tao & Muller (2016) introduced GPT, which has three parts:

- **Gotcha** dense sub-pixel matching;
- a **fourth-order PDE-regularised, total-variation-like MAP** reconstruction;
- a 5 cm output grid, which the authors describe as set heuristically from eight 25 cm passes, with a check of rover wheel-track spacing against Navcam imagery (to within 8 cm) and rock counts.

Later work by the same group moved to learned, GAN-based SRR (e.g. MARSGAN; Tao et al., 2021).

Three questions are rarely answered for such methods:

1. How much of the restored detail is in the data rather than in the prior?
2. How uncertain is each pixel?
3. Can a user trust detail added below the sensor's reach?

Jezero gives an unusual chance to test this: many repeat passes, a compact bright boulder visible in all of them that serves as a blur target, rover photographs of the ground, and Ingenuity's centimetre-scale images of landing hardware.

Contributions:

1. **An open, simplified re-implementation of GPT** (one blur width instead of the original's per-tile 0–1 px choice; no tiling) (MSA-SIFT seeding, adaptive least-squares correlation, best-first region growing, PDE-TV with a measured step bound), with no reference code available.
2. **Per-pass blur measured on a static bright boulder** and transferred to the other passes on static ground.
3. **Noise-weighted, change-masked fusion.** We also show that residual-based pass weighting fails on real HiRISE, because lighting differences between passes masquerade as noise.
4. **A pass-subset bootstrap uncertainty**, calibrated in simulation and reported at the scale where it is honest.
5. **A confidence-scored, measurement-locked detail layer with a user threshold.** Its confidence comes from measured evidence: split-half repeatability, multi-illumination shadows and size. It is calibrated in a simulation whose pass-to-pass disagreement is tuned to real data.

6. **A CPU benchmark against the GPT re-implementation**, with ablations, simulation with exact truth, real split-half consistency and the rover-edge test. Negative results are reported alongside.

2. Related work

- **Repeat-pass SRR for Mars.**
  - GPT (Tao & Muller, 2016) uses Gotcha matching (Otto & Chau, 1989), seeded by SIFT (Lowe, 2004) and refined by ALSC (Gruen, 1985), with a fourth-order PDE regulariser (You & Kaveh, 2000).
  - Learned SRR (e.g. MARSGAN; Tao et al., 2021) produces sharper images whose added texture comes from training data, so measurement and learned prior cannot be separated by the user.
  - We stay with the measurement and add detail only as a separate, locked, confidence-scored layer.
- **PSF estimation.** HiRISE blur varies with spacecraft jitter and focus. We measure it directly on a known target rather than assume one Gaussian.
- **Uncertainty and calibration.** We use the bootstrap (Efron, 1979) over passes; isotonic (Zadrozny & Elkan, 2002) and logistic calibration; and reliability diagrams with expected calibration error (Guo et al., 2017).
- **Ground truth.** Perseverance Navcam (Maki et al., 2020) images were projected to a top-down truth. AI4Mars labels (Swan et al., 2021) give surface classes.

3. Data

- **HiRISE.** 10 HiRISE RDR RED observations of the Jezero landing area, 2007–2024 (`benchmark/data_manifest.json`).
  - Sun incidence is 46–59°.
  - Eight observations have a sun azimuth of 165–178° and are used for restoration.
  - Two (azimuths about 137° and 196°) add illumination diversity for shadow evidence.
- **Test tiles.**
  - A no-rover site at 18.4472 N, 77.4127 E, with 8 passes.
  - A boulder tile at 18.4456 N, 77.4520 E near the landing area, with 3 passes from sols 1–11, containing a compact bright boulder (about 2.5 × 1.8 m) used as the blur and edge target.
- **Rover ground truth.** Perseverance Navcam photos projected to the ground at 50 traverse stops, plus AI4Mars classes.
- **Ingenuity truth.** 190 downward navigation frames (3 cm per pixel from 8 m) of flight 26 (sol 414) over the parachute and backshell, 1.1 km west of the landing spot (Section 6.5).

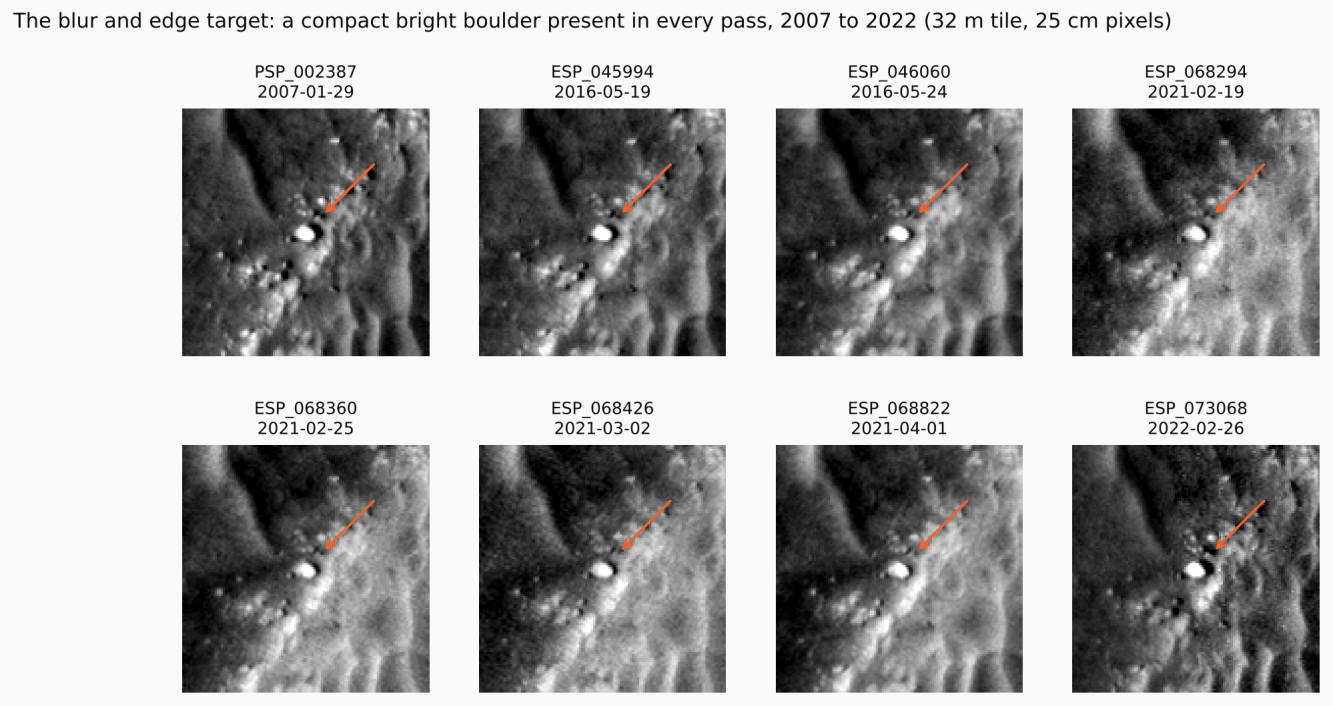


Figure 1. The blur and edge target: a compact bright boulder visible in all 8 passes from 2007 to 2022 (arrow), on the reference grid after Gotcha alignment. 32 m tile, 25 cm pixels, each panel stretched separately.

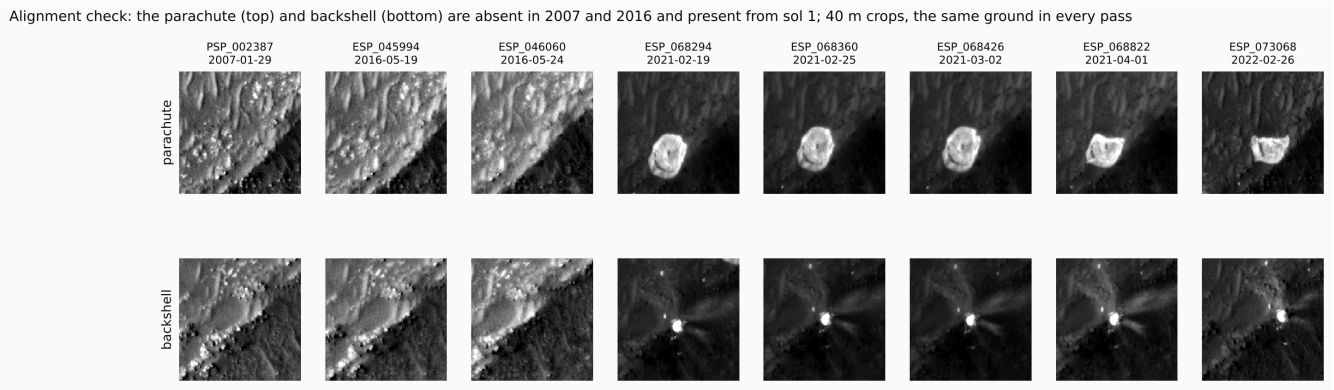


Figure 2. Alignment check over 15 years. The parachute (top) and backshell (bottom) are absent in the 2007 and 2016 passes and present from sol 1, at the same position in every pass. 40 m crops; the parachute's change of shape after March 2021 is visible.

4. Method

4.1 Alignment (Gotcha)

- **Seeds.** SIFT matches, filtered by a ratio test and RANSAC affine; when SIFT fails across years, a phase-correlation seed grid is used instead.
- **Refinement and growth.** Each seed is refined by 8-parameter ALSC (6 affine, gain, offset) and grown best-first on a 4 px grid.
- **Acceptance.** Correlation ≥ 0.6 and a jump ≤ 0.5 px. The rover is masked.
- **Accuracy.** On real texture with a known warp, the median error is 0.038 px (90th percentile 0.089 px), against 0.149 px for tile-wise ECC.

4.2 Restoration (PDE-TV)

- **Energy.** We minimise `E(X) = Σ_k w_k Σ_p m_kp (A_k X − Y_k)_p² + λ Σ_p sqrt(|∇²X|² + μ²)`.
  - `A_k` combines warp, the pass's own Gaussian blur σ_k, and decimation by 4.
- **Solver.** Steepest descent with Nesterov momentum. The step is set from the largest eigenvalue of A^TA, measured by power iteration. The textbook bound underestimates it and diverges.
- **Pass weights.** `w_k` is the inverse white-noise variance of pass k, measured from its finest-scale pixel differences (Immerkær, 1996).
- **Pixel weights.** `m_kp` are Tukey biweights on the residual, re-estimated every 20 iterations. They mask pixels that changed in one pass.
- **Radiometry.** Each pass is gain/offset-normalised to the reference over the aligned overlap.
- **Rejected variant.** Residual-based pass weights, `w_k ∝ 1/σ²(residual)`, are kept as an ablation (Section 6).

4.3 Per-pass blur from the rover

- **Absolute.** On three passes, brightness profiles across the boulder's short axis are fitted with a blurred bar on a linear background. A boulder edge is not a perfect step, so the fitted σ is an upper bound on each pass's blur; the ratios between passes are the robust quantity. With the pixel's box integration removed, `σ = sqrt(s² − 1/12)`.
- **Relative.** The other passes are aligned to the sharpest rover pass on static ground. The extra Gaussian blur that best matches them gives their σ.
- **Results.** ESP_068360 0.59 px (boulder edge), ESP_068426 0.83 px (boulder edge), ESP_068294 0.89 px (boulder edge), ESP_073068 1.16 px (relative), PSP_002387 1.20 px (relative), ESP_045994 1.34 px (relative), ESP_068822 1.38 px (relative), ESP_046060 1.43 px (relative).

4.4 Uncertainty

- **Bootstrap.** The restoration is rebuilt B times from passes drawn with replacement, giving `σ_boot`.
- **Calibration factor.** One factor κ, fitted on one simulated scene and tested on another, absorbs error shared by all passes.
- **Scale.** The uncertainty is reported at the native 25 cm scale.

4.5 Confidence-scored detail

- **Rock probability.** Rock candidates are blob maxima in the restoration. Each gets `P(rock) = σ(w · [z_rep, log D, S·1[D≥1 m], 1[D≥1 m]])`.
  - `z_rep` is the blob response divided by its measurement noise, estimated from two restorations made from disjoint halves of the passes.
  - `S` is multi-illumination shadow evidence: a template correlation per pass, with the shadow on that pass's anti-sun side and length (D/2)·tan(incidence), combined with matched-filter weights.
- **Calibration.** `w` is fitted on simulations whose pass-to-pass disagreement is tuned to real HiRISE (Section 5).
- **Rendering at threshold t.**
  - Rover-photographed rock sprites are added for rocks with P ≥ t.
  - Recogniser-chosen rover texture is added only if t ≤ the class accuracy verified on rover truth.
  - Nothing is added where the base uncertainty exceeds a limit.

    - Every added layer is projected onto the null space of the HiRISE observation.
- **Hazard maps.** Per 1 m cell, the hazard probability is $1 - \Pi(1 - P_i)$ over rocks ≥ 40 cm. A cell is marked unknown where the base is not trusted.

5. Experiments

All experiments run on a 4-core CPU (`benchmark/run_benchmark.py`).

5.1 Methods compared

Method	Description
raw	One pass, bicubic.
previous	Tile ECC + CG Tikhonov.
GPT	Our simplified re-implementation: Gotcha + unweighted PDE-TV, one assumed σ = 1 px, no tiling (the original varies σ between 0 and 1 px per tile).
GPT + weights	Ablation: noise weighting only.
GPT + blur	Ablation: measured per-pass σ only.
full (residual weights)	Ablation: the first weighting variant.
full	This work.
oracle	Full with exact motion; simulation only.

5.2 Tests

- **A. Simulation with exact truth.**
    - 5 scenes of fractal ground.
    - 8 passes with the measured blurs, a random offset plus 0.3 px sinusoidal jitter, 3% noise, and a 2 m change planted in one pass.
    - Every method estimates its own motion.
- **B. Real split-half consistency.** The 8 real passes are split into disjoint halves, each restored independently. Fourier-ring coherence between the halves measures detail that repeats in independent data.
- **C. Boulder edge.** The edge width of the bright boulder in each restoration, with ringing (RMS residual of the bar fit) and the largest undershoot beside the edge.
- **D. Uncertainty calibration.** As Section 4.4.
- **E. Confidence calibration and the real-data check.**
    - The simulation's per-pass noise and illumination mismatch are tuned so its split-half disagreement matches the real tile at every rock size. Best setting: noise 0.50, illumination 0.60; log-ratio cost 0.019.
    - Then 10 training and 6 held-out scenes.
    - Real check: the same scores against the rover answer key, with a shifted-truth control.

6. Results

6.1 Simulation (test A)

Method	Error, 25 cm (%)	Error, 6.25 cm (%)	Artefact at change (× std)	Resolved (cm)
Raw (1 pass, bicubic)	12.82	38.71	1.26	40
Previous (ECC + CG Tikhonov)	5.16	34.83	1.48	40
GPT (our re-implementation)	3.55	30.20	1.43	33
GPT + noise weighting	4.68	32.58	1.09	33
GPT + measured per-pass blur	2.60	27.64	1.29	29
Full, residual weights (ablation)	1.44	29.86	1.03	33
Full (this work)	1.43	29.78	1.01	33
Full, exact motion (upper bound)	1.25	29.81	1.00	33

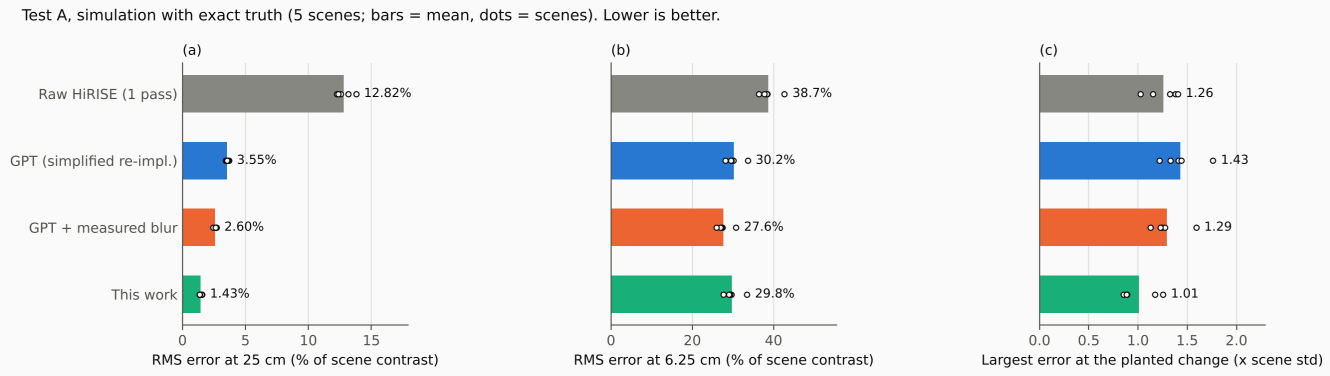


Figure 3. Test A, simulation with exact truth. (a) RMS error at the native 25 cm scale; (b) at the 6.25 cm output pixel; (c) the largest error at a change planted in one pass. Bars are means over scenes and dots are individual scenes; lower is better.

Paired against GPT, by scene:

- **25 cm:** -2.13 (95% CI -2.21 to -2.02), better in 5/5 scenes.
- **Artefact at the planted change:** -0.42 (95% CI -0.50 to -0.34), better in 5/5 scenes.
- **6.25 cm:** -0.42 (95% CI -0.53 to -0.29), better in 5/5 scenes.
- **Measured blur.** Measured per-pass blur alone moves GPT from 3.55% to 2.60% at 25 cm.
- **Noise weighting.** Noise weighting alone gives 4.68%.
- **Residual weights.** The residual-weighted variant reaches 1.44% in simulation; see 6.2 for why it is rejected.
- **Alignment headroom.** With exact motion the full chain reaches 1.25%, so alignment leaves little to gain.

6.2 Real HiRISE (tests B and C)

Split-half consistency (8 real passes; halves PSP_002387, ESP_045994, ESP_046060, ESP_068426 vs ESP_073068, ESP_068294, ESP_068360, ESP_068822). Higher is better.

Coherence above 0.5 at fine scales for the previous pipeline is a reconstruction-grid artefact shared by both halves, not detail.

Method	Halves agree down to	Coherence at 100 / 67 / 50 cm	Mean coherence, 25–100 cm
Raw (1 pass, bicubic)	100 cm	0.75 / 0.44 / 0.07	0.183
Previous (ECC + CG Tikhonov)	67 cm	0.87 / 0.70 / 0.26	0.425
GPT (our re-implementation)	67 cm	0.87 / 0.64 / 0.18	0.263
GPT + noise weighting	67 cm	0.84 / 0.62 / 0.18	0.271
GPT + measured per-pass blur	67 cm	0.86 / 0.59 / 0.10	0.225
Full, residual weights (ablation)	67 cm	0.80 / 0.50 / 0.07	0.199
Full (this work)	67 cm	0.83 / 0.60 / 0.11	0.230

Boulder edge (bright static boulder near the landing area, 3 passes, sols 1–11). A narrower edge is sharper. Lower ringing and undershoot are cleaner.

Method	Edge σ (cm)	Ringing, RMS (% of step)	Largest undershoot (% of step)
Raw (1 pass, bicubic)	23.3	9.2	18.1
Previous (ECC + CG Tikhonov)	20.6	10.4	20.9
GPT (our re-implementation)	14.1	16.2	42.2
GPT + noise weighting	15.6	16.1	37.1
GPT + measured per-pass blur	14.9	11.6	30.0
Full, residual weights (ablation)	16.4	12.6	25.2
Full (this work)	16.7	11.7	25.6

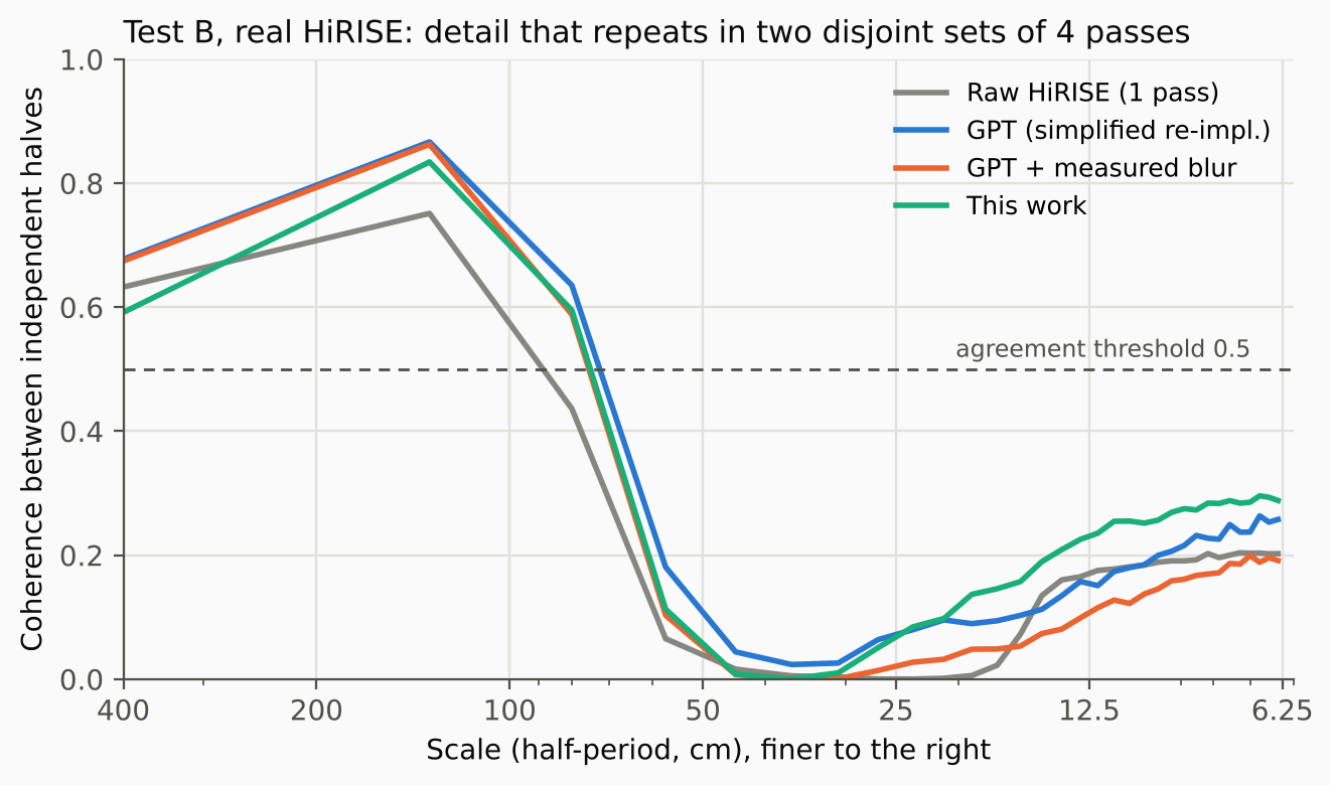


Figure 4. Test B, real HiRISE. Coherence between restorations made from two disjoint sets of 4 passes, against scale. Detail is treated as repeatable where coherence is at least 0.5 (dashed line).

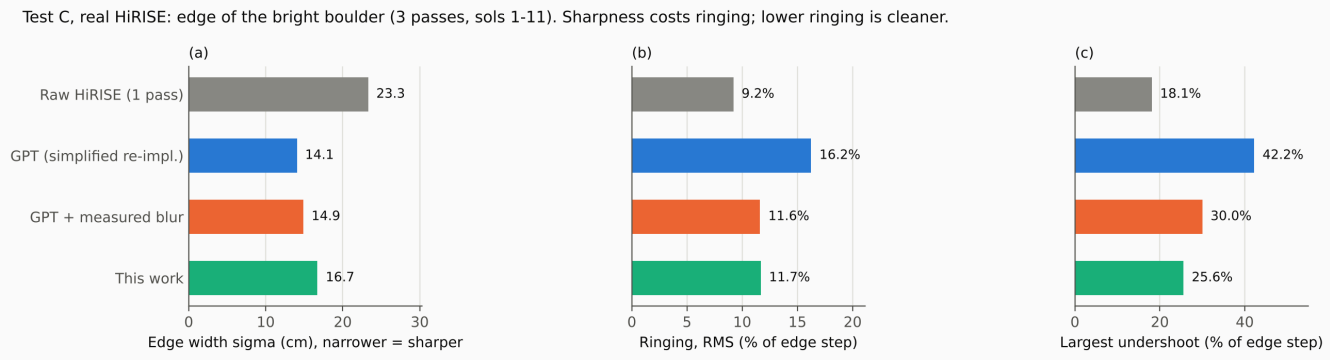


Figure 5. Test C, real HiRISE, the bright boulder's edge. (a) Fitted edge width; (b) RMS ringing; (c) largest undershoot, both as a share of the edge's brightness step.

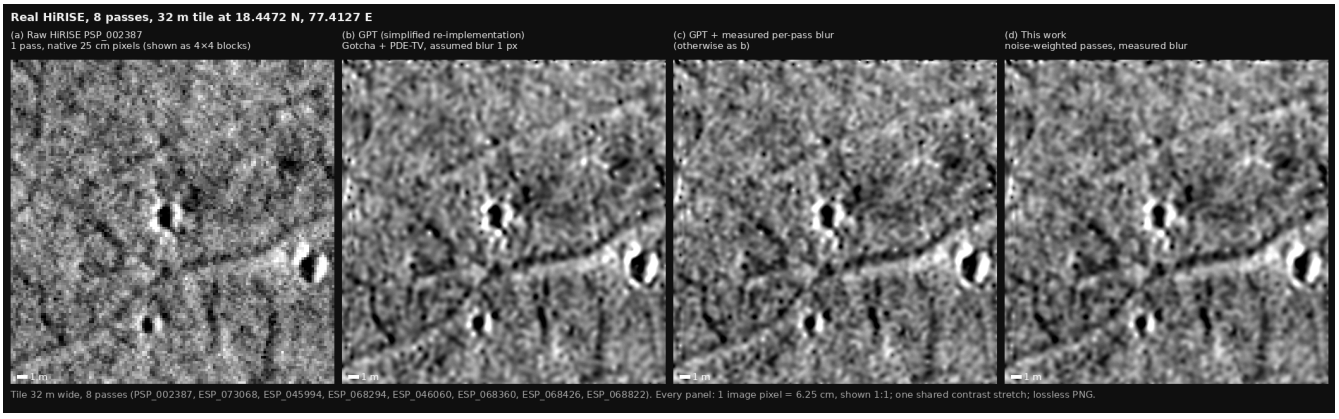


Figure 6. Real HiRISE, 8 passes. (a) One raw pass with its 25 cm pixels shown as blocks; (b) GPT (simplified re-implementation); (c) GPT with measured per-pass blur; (d) this work. 6.25 cm output pixels shown 1:1, with one shared contrast stretch.

- **Split-half consistency.** Mean coherence over 25–100 cm is 0.230 for the full chain and 0.263 for GPT (GPT ahead).
- **Boulder edge.** The edge is 16.7 cm (full) against 14.1 cm (GPT).
  - Ringing: 11.7% against 16.2%.
  - Undershoot: 25.6% against 42.2%.
- **Residual weights.** The residual-weighted variant reaches 0.199 on split halves. Residual weights read illumination differences between passes as noise and collapse onto one pass, so each half is effectively a single image.

6.3 Uncertainty (test D)

Scale	κ	Held-out coverage ±1σ / ±2σ	Spearman(σ, |err|)	Error, least vs most confident 10%
6.25 cm (output pixel)	12.8	81.5 / 96.0%	0.14	25.7 vs 21.6%
12.5 cm	4.8	85.1 / 97.9%	0.34	9.5 vs 4.5%
25 cm (native HiRISE pixel)	2.1	86.4 / 99.1%	0.39	3.8 vs 0.6%

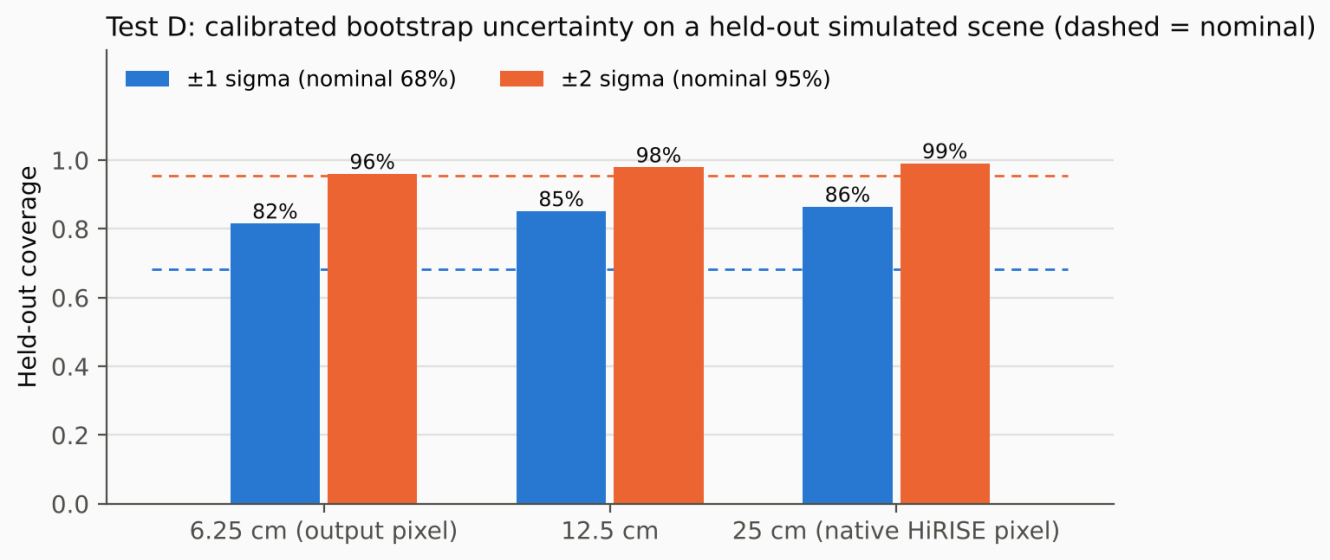


Figure 7. Test D. Held-out coverage of the calibrated bootstrap uncertainty at three scales; dashed lines are the nominal 68% and 95%.

6.4 Confidence (test E)

- **Held out (3692 candidates, 648 real rocks):** AUC 0.70, ECE 0.4%.
- **Without the shadow feature:** AUC 0.71.
- **AUC by size:** 25 cm 0.61, 60 cm 0.76, 100 cm 0.55, 150 cm 0.71.

Threshold	Shown	Precision	Recall	Recall by size (25/40/60/100/150 cm)
0.1	554	19%	16.0%	1 / 2 / 15 / 60 / 92
0.2	157	30%	7.3%	0 / 0 / 2 / 22 / 65
0.3	67	42%	4.3%	0 / 0 / 2 / 13 / 38
0.4	31	55%	2.6%	0 / 0 / 1 / 5 / 27
0.5	19	53%	1.5%	0 / 0 / 1 / 0 / 19
0.6	8	50%	0.6%	0 / 0 / 0 / 0 / 8
0.7	3	67%	0.3%	0 / 0 / 0 / 0 / 4
0.8	2	50%	0.2%	0 / 0 / 0 / 0 / 2

Real-data check.

- Against the traverse rover truth, candidate positions match truth rocks 29% of the time, against 18% with the truth shifted 40 px. So the positions carry signal.
- But no score ranks candidates: AUC 0.43 for the physical score and 0.46 for the recogniser.
- The recogniser's surface class is right 26.6% of the time, against 49.9% for always guessing soil.
- Only 4 of 27 stops could be registered, with shifts of up to 2–3 m. The answer key, not the scores, is the limit.

6.5 Real physical truth: Ingenuity's view of the backshell

Ingenuity photographed the debris field from 8 m (flight 26, sol 414) at about 3 cm per pixel. A mosaic of 27 frames around the backshell was placed in HiRISE by (i) the helicopter's own shadow and the computed sun (mosaic-to-north rotation 217 deg +- 12 deg) and (ii) masked correlation against three independent HiRISE passes (208 deg, scale 0.0290 m/px = the nominal 8 m / 275 px, the same translation in all three). The 181 m2 rock field around the backshell gave no discrimination between methods (correlation 0.00-0.07 for all, including raw HiRISE), because the truth (midday) and HiRISE (afternoon) show rocks with completely different shading. Within 4 m of the backshell the two share strong structure (correlation 0.92-0.97 at 1-4 m), so the test below uses that object; it was chosen after the rock-field result and carries the safeguards listed. Out of sample (lambda chosen on one half of the disc, scored on the other): +0.001 (1 m-50 cm) and +0.009 (50-25 cm). The 1 m-50 cm advantage is stable across disc radii of 2.5-8 m (+0.007 to +0.010 in correlation); the 25-50 cm advantage is not (it disappears at 8 m). Over 45 plausible placements the full chain is ahead of GPT in 91% of them at 50-25 cm and 84% at 1 m-50 cm.

Error = 1 - correlation with the truth in the 25-50 cm band, within 4 m of the backshell, with a +-0.25 m placement tolerance applied equally to all methods.

Method	Error at its minimum (50-25 cm)	at lambda	at lambda = 0.005	at lambda = 0.04
GPT (simplified re-implementation)	0.229	3.0	0.368	0.333
GPT + measured blur	0.227	3.0	0.296	0.286
this work, equal weights	0.214	0.1	0.281	0.234
this work	0.203	0.1	0.248	0.206
raw HiRISE	0.249	none	none	none

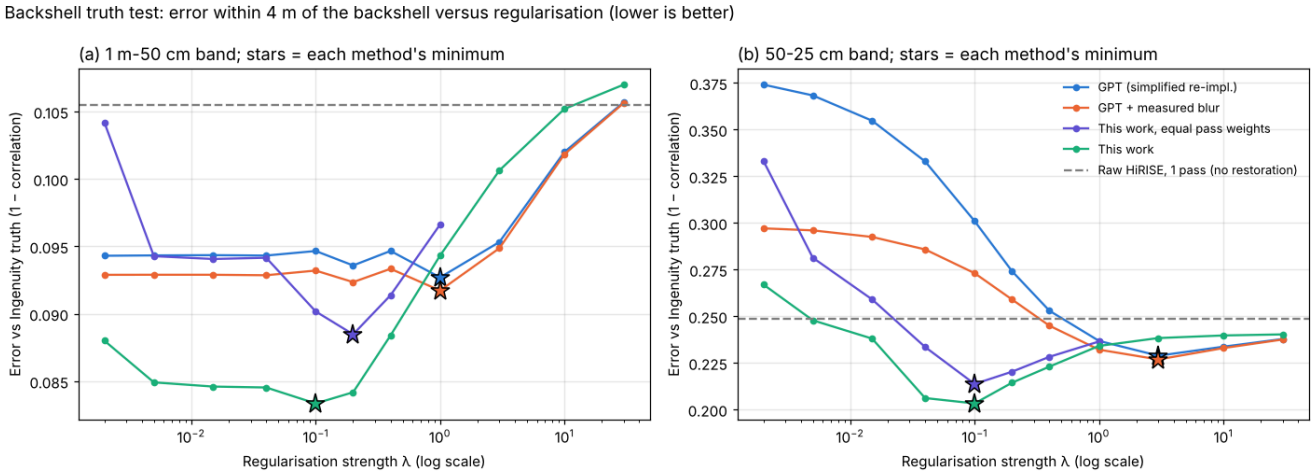


Figure 8. Backshell truth test: error against Ingenuity's view within 4 m of the backshell, versus regularisation strength, in (a) the 1 m–50 cm band and (b) the 50–25 cm band. Stars mark each method's minimum; the dashed line is raw HiRISE.

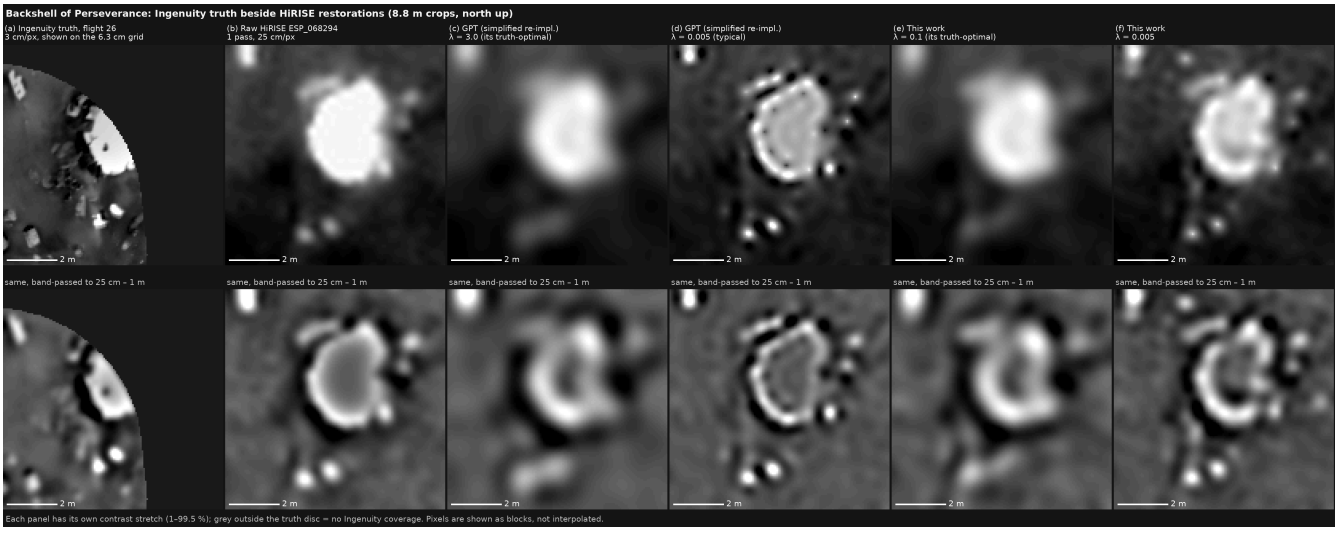


Figure 9. The backshell neighbourhood: (a) Ingenuity truth beside (b) raw HiRISE and (c–f) restorations at each method's truth-optimal and typical λ. The bottom row is band-passed to 25 cm–1 m, the scales the test scores.

6.6 Shadow evidence

AUC for rock against decoy (a flat bright patch) / rock against ground:

Setup	40 cm	60 cm	100 cm	150 cm
1 pass	0.52 / 0.71	0.48 / 0.82	0.86 / 0.98	0.88 / 0.99
8 same-sun passes	0.56 / 0.62	0.53 / 0.78	0.90 / 0.99	0.86 / 0.99
10 passes incl. 2 different sun angles	0.55 / 0.65	0.52 / 0.80	0.90 / 0.99	0.86 / 0.99

7. Discussion

Scope of the claims. We claim a measured improvement at the 25 cm scale in simulation and on real HiRISE, a small improvement at one physical object, calibrated uncertainty at 25 cm, and a reproducible protocol for testing restoration against independent truth. We do not claim

detail below the scales reported in Section 6, and our baseline is a simplified GPT (Section 5.1), so the comparison is with that baseline and not with the original software.

- **What the gain is.** The improvement over GPT comes from measuring what GPT assumes:
  - the blur of each pass;
  - which passes are noisy;
  - which pixels changed.

  It does not come from a stronger prior.
- **Assumed blur sharpens and rings.** In our baseline the assumed blur (1 px) is larger than the boulder-measured blur of the sharp passes, so on real data it deconvolves harder. The original GPT chooses σ between 0 and 1 px per tile, which may reduce this effect; we did not test it. That sharpens edges and produces ringing (Section 6.2). Edge sharpness alone therefore rewards a wrong blur model, and ringing should be reported with it.
- **Residual weighting collapses on real data.** Weights based on residuals work in simulation but collapse onto a single pass on real data. Real passes differ by illumination and season far more than by noise. Noise must be measured from the pass itself.
- **No 5 cm claim.** On real HiRISE, detail that repeats in independent passes stops at about 67 cm; even in simulation with exact truth, the finest resolved scale is about 33 cm. Below that, a restoration's content comes from the prior. Tao & Muller (2016) report 5 cm on eight passes over the Gusev crater traverse; we did not test that site, and our evidence is limited to the sites and passes above. We do not claim 5 cm, and we do not claim to refute it.
- **Where confidence is limited.** Confidence below about 60 cm cannot be verified with current rover truth. Our confidences are therefore simulation-calibrated, with a simulation whose disagreement is tuned to real data. The real-data check confirms positional signal but cannot confirm the probabilities.
- **A physical truth at one object.** Ingenuity's images, placed in HiRISE by two independent methods (a sun-and-shadow compass and image correlation on three passes), give a centimetre-scale truth for the backshell. It supports a clear advantage over the GPT re-implementation at any fixed regularisation and a small advantage once each is tuned (Section 6.5), but it is a single object. A wider validation needs truth with several metre-scale shared structures.
- **What would close the gap.** Better-registered truth would: more shared hardware or tracks as tie points, or stereo from rover-mast cameras.

8. Conclusion

Measuring each pass's blur on a static target and weighting passes by their own noise gives a measurable gain over GPT at the native 25 cm scale in simulation and on real HiRISE, and a small gain at one physical object. A confidence-scored, measurement-locked detail layer lets users decide how much unverified detail to see, with probabilities that are calibrated where truth exists and labelled where it does not.

Reproducibility

- **Code and tests.** All code, tests (`pytest`, run on GitHub CI) and result files are in this repository. Raw data (about 12 GB) is listed with URLs in `benchmark/data_manifest.json`.
- **CPU time (4 cores).**
  - Benchmark part A: 1208 s.
  - Part B: 410 s.
  - Part C: 148 s.
  - Bootstrap on a 32 m tile with 8 passes: 697 s.

References

Checked against the paper itself or publisher records: Tao & Muller (2016; read in full), Tao et al. (2021), Maki et al. (2020), Swan et al. (2021). The remaining entries are standard references cited from memory with volume and page numbers that we believe correct; confirm them against the publishers before submission.

- Efron, B. (1979). Bootstrap methods: another look at the jackknife. *Ann. Statist.* 7, 1–26.
- Fischler, M. A., Bolles, R. C. (1981). Random sample consensus. *Commun. ACM* 24, 381–395.
- Gruen, A. (1985). Adaptive least squares correlation. *S. Afr. J. Photogramm. Remote Sens. Cartogr.* 14, 175–187.
- Guo, C., Pleiss, G., Sun, Y., Weinberger, K. Q. (2017). On calibration of modern neural networks. *ICML*.
- Immerkær, J. (1996). Fast noise variance estimation. *Comput. Vis. Image Underst.* 64, 300–302.
- Lowe, D. G. (2004). Distinctive image features from scale-invariant keypoints. *IJCV* 60, 91–110.
- Maki, J. N., et al. (2020). The Mars 2020 Engineering Cameras and Microphone on the Perseverance Rover: A Next-Generation Imaging System for Mars Exploration. *Space Sci. Rev.* 216, 137. doi:10.1007/s11214-020-00765-9
- McEwen, A. S., et al. (2007). Mars Reconnaissance Orbiter's High Resolution Imaging Science Experiment (HiRISE). *J. Geophys. Res.* 112, E05S02.
- Otto, G. P., Chau, T. K. W. (1989). 'Region-growing' algorithm for matching of terrain images. *Image Vis. Comput.* 7, 83–94.
- Swan, R. M., et al. (2021). AI4MARS: A dataset for terrain-aware autonomous driving on Mars. *CVPR Workshops*, 1982–1991. doi:10.1109/CVPRW53098.2021.00226
- Tao, Y., Muller, J.-P. (2016). A novel method for surface exploration: super-resolution restoration of Mars repeat-pass orbital imagery. *Planet. Space Sci.* 121, 103–114 (GPT; 5 cm demonstrated on eight HiRISE passes over the Spirit traverse in Gusev crater).
- Tao, Y., Conway, S. J., Muller, J.-P., Putri, A. R. D., Thomas, N., Cremonese, G. (2021). Single image super-resolution restoration of TGO CaSSIS colour images: demonstration with Perseverance rover landing site and Mars science targets (MARSGAN). *Remote Sens.* 13(9), 1777. doi:10.3390/rs13091777
- You, Y.-L., Kaveh, M. (2000). Fourth-order partial differential equations for noise removal. *IEEE Trans. Image Process.* 9, 1723–1730.
- Zadrozny, B., Elkan, C. (2002). Transforming classifier scores into accurate multiclass probability estimates. *KDD*.